

Know-your-own-league context: insights for player preparation and recruitment - Part 1: Team formations

Martin Buchheit,^{1 2 3} Maxime Settembre,¹ Adrien Tarascon,² Karim Hader,¹ Adam Stokes,¹ Allan Munro,¹ Derek McHugh,¹ Raymond Verheijen⁴

¹Kitman Labs, Dublin, Ireland

²Lille OSC, Lille, France

³HIITscience, Revelstoke, Canada

⁴Football Coach Evolution, The Netherlands

Soccer | Association Football | System of Play | Player Interaction | Coaching | Tactics

Headline

Football is a total social phenomenon (4), i.e., one of the only phenomena, if not the only one, that involves all the societal institutions - social, cultural, economical and political. There are more than 250 million participants worldwide. While it's the same sport that is played across all countries and leagues, with a very large proportion of foreign players, coaches and general managers in each league, there is a common belief that the sport may be played differently in each country (13). Strong cultural heritage, stereotypes, e.g., English fighting spirit, Spanish toughness, French flair, German efficiency, and likely visions concerning the style of play and game approach may explain these differences (3,11, 13). Given the increasingly globalised game, understanding how football is played in each league can add context to a team's foreign player recruitment process, especially considering team formations (for/against), or to better develop players that will ultimately play in these leagues. It can also provide a more thorough understanding for coaches preparing matches in European competitions like the Champions League. There is limited research available around between-league differences in playing style (11, 13), typical team formations, player management, e.g., player turnover (10), match-to-match rotations and substitutions, results (percentage of home vs. away wins and percentage of draws), number of goals scored (1) (home vs. away), and the number of rotations (home vs. away) - with the majority of studies focusing on typical in-game metrics, e.g., number of passes and shots on goals (5,7,17).

The first part of our series of short papers focuses on team formations and the evolution of their use over the past 15 years in the top seven European leagues, i.e., the English Premier League (EPL), the French Ligue 1, the German Bundesliga, the Dutch Eredivisie, the Italian Serie A, the Portuguese Liga and Spanish Liga.

Aim

The first aim of this study was to examine between-league differences in typical team formations, and how these may have changed over the years in each league. We also looked at how often teams change formations within a season, what the typical adjustments made in relation to the opponent's formation are, and whether all of this is affected by match context, i.e., home vs. away.

Methods

Data Extraction

We extracted the fixtures data of teams competing in the top seven European leagues, i.e., the English Premier League

(EPL), the French Ligue 1, the German Bundesliga, the Dutch Eredivisie, the Italian Serie A, the Portuguese Liga and Spanish Liga. This includes 15 seasons from 2007/08 to 2021/22. The fixtures analysed included all domestic League competitions. This data was sourced from Transfermarkt. Across the 15 seasons, this represents 252 teams and more than 35,000 fixtures. While there was more data available on-line (i.e., from early 2000), our data health checks indicated that the reporting of team formations wasn't always consistent across all leagues up to, and including, the 2006/07 season. The information was missing for between 60% to 95% of fixtures within most leagues, mainly impacting the weaker teams. As the data was not missing at random, team formation analysis was therefore restricted to the 2007/08 - 2021/22 period.

Definitions

Football formations are usually set up by managers or coaches to help players in the team act in an organised manner and know their roles. In fact, improving the communication between players is likely the most important aspect of performance (15). The six most frequent and representative team formations were first defined by a pool of analysts working in Champions League clubs: 3-4-3, 3-5-2, 4-3-3, 4-4-2, 4-4-2 diamond and 4-5-1. When a given team formation was not one of these six common formations, the same pool of analysts classified them within the closest summary formation available (Table 1). It is worth noting that there is no "right" way to summarise these formations. The granularity has changed over the years where formations like 3-4-2-1 and 4-3-2-1 have become more common recently. Taking a step back, we are attempting to use the formation as a proxy for how a team is set up tactically, knowing that this encoding cannot capture how a team reacts offensively and defensively over 90 mins. However, the formation does contain information about the manager's intent and we have tried to capture this in a general way in how we have grouped the formations.

Data analysis

We studied the proportion of formations at different levels in order to understand the different factors related to the decision behind their choice:

- per season for all leagues pooled together (Figure 1)
- per season and per league (Figures 2 to 7). For the two above cases, we fitted a least-squares approximation with 95% confidence intervals for the estimate. Our choice of

this approximation was to get a sense of the main tendency over the entire period of time, on top of capturing seasonal differences.

At league level, we also calculated the proportion of formations in order to analyse:

- the impact of playing home or away (Figure 8)
- the use of 3-defender or 4-defender formations depending on the opponent (Figure 9)
- the consistency of formation usage by teams (Figures 10 and 11). Lastly, we focused on the tactical evolution of some of the top teams in Europe (Figures 12 and 13).

All league graphs are ranked based on the UEFA index on 26/10/2022.

Table 1. Results

Summary Formation	Observed Formation
3-4-3* (6%)	3-4-2-1 3-4-3 5-4-1
3-5-2 (8%)	3-1-4-2 3-4-1-2 3-5-2 3-5-2 Attacking 3-5-2 Flat 5-3-2
4-3-3 (31%)	4-1-4-1 4-3-2-1 4-3-3 4-3-3 Attacking 4-3-3 Defending 4-3-3 Sweeper
4-5-1 (28%)	4-2-3-1 4-5-1 4-5-1 Flat
4-4-2 (19%)	4-4-1-1 4-4-2 4-4-2 Double 6 4-4-2 Sweeper
4-4-2 Diamond (8 %)	4-1-3-2 4-3-1-2 4-4-2 Diamond 4-4-2 Sweeper Diamond

Results and Discussion

This study is the first to our knowledge to examine between-league differences in team formations and player management, namely match-to-match player rotations and substitutions.

The main limitation of the current analysis remains the team formations data set obtained from an online database. In particular, there is always a degree of uncertainty in the way a formation is recorded on a public website like Transfermarkt relative to what was actually registered. Furthermore, how a team is actually organised on the pitch can be quite different from the registered formation on paper (generally at kick-off in a defensive mid-block). Additionally, today teams tend to vary their formation during matches to adapt to the context,

especially in defensive vs. offensive phases. As there is some consistent variability of formations within single matches, trying to be too granular in terms of formations may not only be impossible (e.g., information not provided and subject to interpretation), but may not make much sense either given the fluidity of modern football. Finally, all available formations were grouped into six categories to simplify the analysis (Table 1), and whether a given formation belongs to one of our categories or another will be hotly debated among practitioners. We nevertheless believe that the global trends described remain of interest, and would not be substantially impacted by those potential limitations.

Overall between-league differences

The first finding of the present study is that there were some clear between-league differences in terms of preferred team formations. Globally, the 4-3-3 (31%), the 4-5-1 (28%) and the 4-4-2 (19%) were the most used systems over the 15-year period examined (Table 1). This was consistent at the domestic level for some leagues, such as in the Eredivisie (consistently 70% of use throughout the years) and Liga Portugal (60% to 30% of use), where the 4-3-3 has been the most popular formation since the first season examined (Figures 2 and 6). It is worth noting also that the “culture of the 4-3-3” is really outstanding in the Eredivisie since it is the only league where a single team formation has dominated that much over the others (Figure 2). In contrast, for example, the use of the 4-3-3 remained constantly as low as 15% in the Bundesliga (Figure 2). Other formations were also more dominant in some leagues, e.g., in the Serie A (Figures 5 and 6), coaches were the pioneers in using 3-defenders formations, which were as popular as the three main formations (Table 1).

The fact that main team formations were not uniformly used across all leagues, especially over the first decade examined (see next section), as in the Eredivisie (4-3-3) and Serie A (“back three”), shows that context and culture still had some influence on tactical decisions, and may be relevant for both scouts, individual player development, professionals and coaches working/preparing matches across multiple leagues. For example, except in Serie A, the majority of teams had to build their offensive phases towards the attack of a line of four defenders. Strikers and offensive players may also need to develop specific individual skills to be more successful against these defensive systems. This league specificity was likely related to the fact that in many leagues, the majority of coaches were native to the league country (e.g., Serie A, Ligue 1), which tends to extend this cultural influence. It is however possible that the more multinational nature of modern football has progressively wiped off these leagues’ differences; the EPL being the first league that became international at the player, coach or general manager levels. Whether this development over the years has actually diluted this domestic culture is difficult to examine within a single league (one change may compensate the other), but is well reflected as discussed in the following section by 1) the homogenisation of team formations within most of the leagues in the past 5-6 years (Figure 6) and 2) the high variability of teams formations used in single clubs (Figures 12 and 13).

Overall trends across the years

The second main finding around team formations is that there was a trend over time towards increasing use of 3-defender formations (i.e., 3-4-3, 3-5-2) at the expense of the 4-4-2 and 4-5-1 (Figure 1). There were nevertheless some important between-league differences, and zooming in on the evolution of these

different formations over time provided some interesting insights that are discussed below.

Regarding the 4-3-3 (Figure 2), while the use remained pretty constant over time in some leagues (Bundesliga, Serie A, Ligue 1, Eredivisie), there were important decreases (Liga Portugal, from 60% to 30%) or increases in the use of that system in others (La Liga).

The drop in the use of the 4-4-2 over the time, shown globally in Figure 1 happened essentially before the 2012/2013 season and remained more stable onwards (Figure 3, upper panel). This trend was consistent across many leagues and especially the EPL where the relative use decreased from 70% in 2007/2008 to only 20% in 2012/2013 (Figure 3). However, it is worth noting that this dynamic was inverted in La Liga and Liga Portugal, where there was rather an increased use of the 4-4-2 after 2012/2013. The drop in the use of the 4-4-2 Diamond shape paralleled perfectly the overall decrease in the use of 4-4-2 to the point of almost not being used anymore (Figure 1), which is consistent with some recent observations (12).

The overall decreasing trend of the 4-5-1 over time described in Figure 1 appears in fact essentially related to the decrease

of use in La Liga (Figure 4); the relative use remained more or less stable in the other leagues, with even an increase in the Liga Portugal from 2012/2013.

Overall, the dynamic evolution of the 4-4-2 and 4-5-1 over the years suggests that their performance and usefulness have been questioned year after year, and trial and error phases may have in turn led to a progressive decrease in their use across all leagues (around 40% of use today for both systems in all leagues vs. > 60% 20 years ago). This decreased use of the 4-4-2 and 4-5-1 team formations occurred at the benefit of 3-defender team formations (i.e., 3-4-3 and 3-5-2, Figures 1 and 5). This clear increase in the use of a “back three” shown in Figure 1 is actually reflective of a trend that occurred in all leagues (Figures 5 and 7) - and this is the only trend that was really consistent across all leagues (e.g., the decreased use of the 4-4-2 observed globally was not observed in La Liga and the Liga Portugal, Figure 3). The main between-league difference remains in the fact that while this increase of “the back three” kicked in during the 2014/2015 season for most teams, these formations were already important in the Serie A since 2007/2008 (i.e., beginning of our analysis).

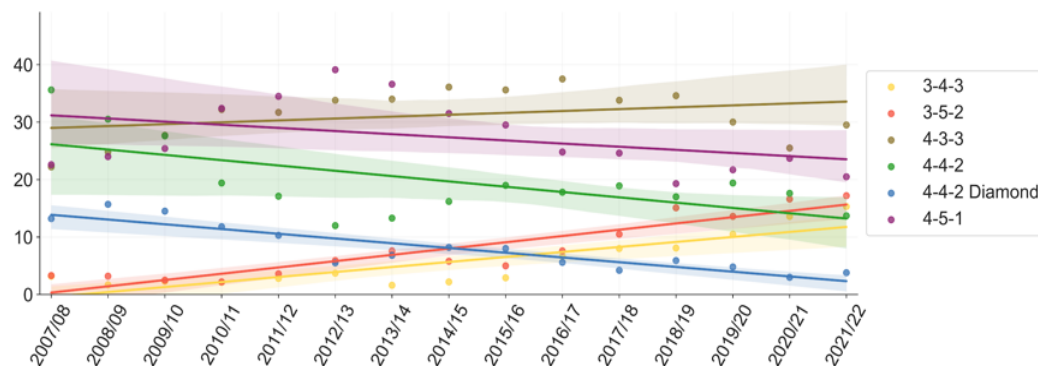


Fig. 1. Proportion of team formations for all leagues pooled together from 2007/08 to the 2021/22 seasons.

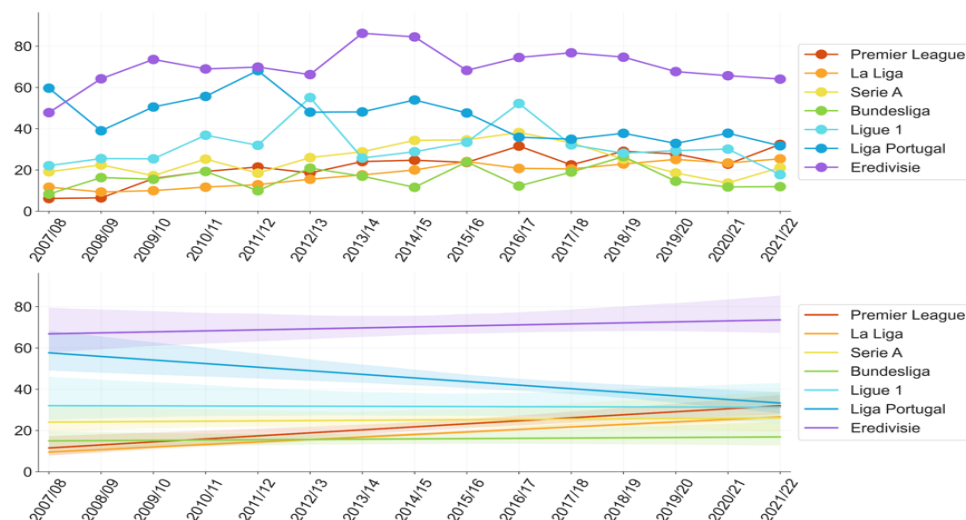


Fig. 2. Proportion of 4-3-3 team formations (upper panel) and linear trends for the proportion of 4-3-3 team formations (lower panel) for the seven leagues individually from the 2007/08 to the 2021/22 seasons.

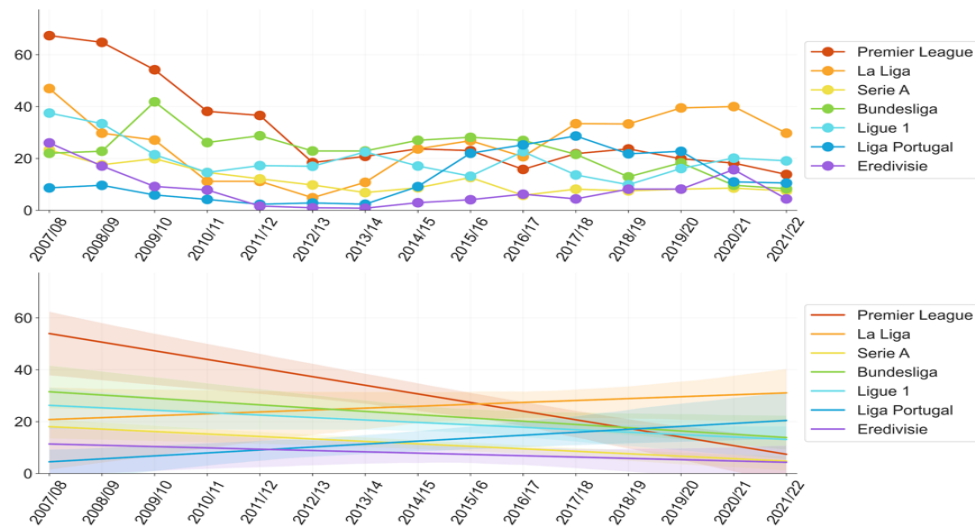


Fig. 3. Proportion of 4-4-2 team formations (upper panel) and linear trends for the proportion of 4-4-2 team formations (lower panel) for the seven leagues individually from 2007/08 to the 2021/22 seasons.

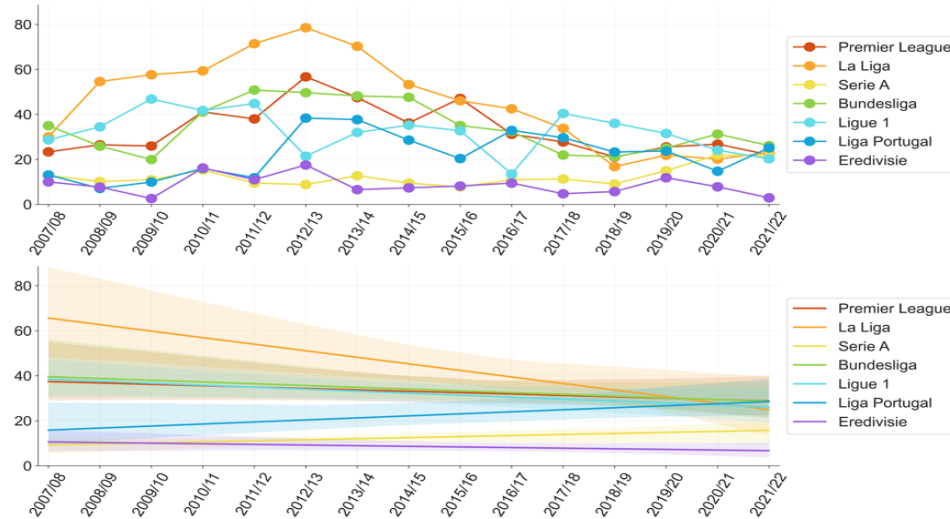


Fig. 4. Proportion of 4-5-1 team formations (upper panel) and linear trends for the proportion of 4-5-1 team formations (lower panel) for the seven leagues individually from 2007/08 to the 2021/22 seasons.

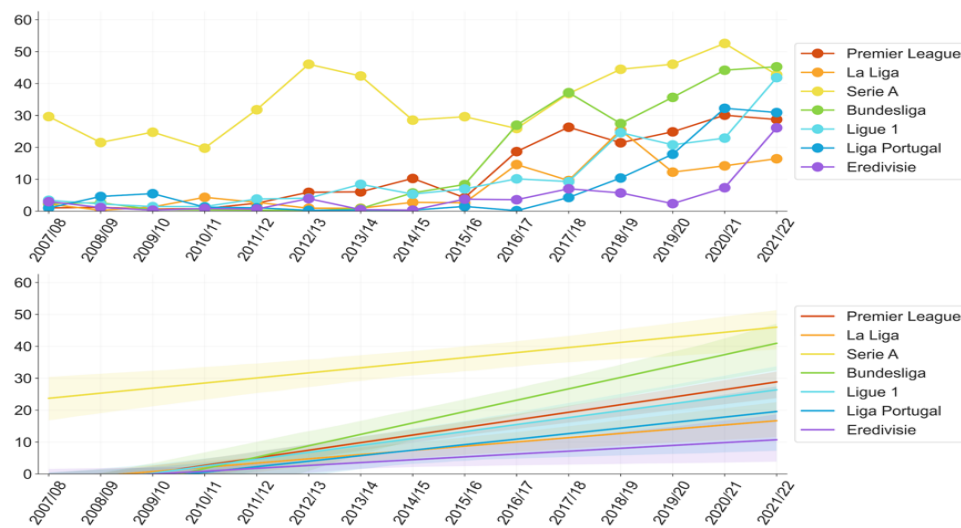


Fig. 5. Proportion of 3-defenders (3-4-3 and 3-5-2 pooled together) team formations (upper panel) and linear trends for the proportion of these team formations (lower panel) for the seven leagues individually from 2007/08 to the 2021/22 seasons.

Specific periods across the years

When looking at leagues more individually (Figures 6 and 7), some between-league differences were accentuated at some specific moments in the history; for example, the use of the 4-5-1 ranged from 80% in La Liga vs. 10% in the Serie A in 2012/2013 (Figure 4). In Ligue 1, the proportion of use of the 4-4-2, 4-5-1 vs. 4-3-3 was constantly changing so that 4-3-3 even dominated over some short periods of time (e.g., 2012-2013 and 2016/2017). In the EPL, Bundesliga, Ligue 1 and Serie A, the use of the 4-4-2 decreased almost linearly until today (Figure 6). In La Liga however, after a similar linear decrease from 60% to almost 0% of use in 2012/2013, there has been a linear increase that reached 40% in 2017/2018 and remained stable until today. In most of the leagues, except in the Eredivisie, the drop in the use of the 4-4-2 coincided with the rise of the 4-5-1, especially from the 2009/2010 season (Figure 6). Interestingly, this rise of the 4-5-1 (more precisely the 4-2-3-1) was also reflected in the team formations mainly used during the 2010 World cup (Barcelona and Spain leading the show) (16). However, for four to six years the 4-5-1 has now also started to be abandoned (especially in EPL, Liga, Ligue 1 and Bundesliga), to leave more space for formations with three defenders.

In many leagues, a given team formation was often preferred over the others for a large number of consecutive years, before being replaced by another (Figure 6): in the EPL, the 4-4-2 and 4-5-1 were clearly more used for a period of nine years; in La Liga, the 4-5-1 dominated for a block of nine years; in the Portuguese Liga, the 4-3-3 dominated for seven years; and in the Bundesliga, it's the 4-5-1 that was more used for seven years. In contrast, in the French Ligue 1 and the Serie A, the preferred formations kept changing over the years. If we now look at the last five seasons, with the exception of the Eredivisie, there seems to be a uniformisation of team formations in all leagues, with less and less tendency for dominance of a single formation over the others (Figure 6). The similarities between leagues in terms of trends over the years, and especially the rise of the “back three” (Figure 5) are consistent with previous findings (7) and perhaps highlight an increasing homogenisation of modern football during these years through the increased movement of staff and players between leagues,

who each time bring their own culture and context to their new club, which tends to progressively decrease local influences. This may also be related to the fact that many of the top managers follow the same coaching courses (e.g., UEFA coaching licence), and may tend to copy what their peers do to avoid losing any possible innovation advantage; overall, this tends to standardise the tactical approach of matches.

Effect of match location

When examining the effect of match location on team formations (Figure 8), there was a trend for a more defensive approach to the away matches, which is consistent with the home advantage theory (6,9). For example, there was a strong trend across almost all leagues and years for greater use of 3-defender formations when away (Figure 8) (which, added to the 4 to 5 midfielders that can drop when losing the ball, increases players' density in defence). While not as consistent as the latter, the other noticeable trend (especially before 3-defenders formations started to be used, <2013/2014), was to play more often away with 5 rather than 4 midfielders (4-5-1 > 4-4-2). Interestingly, despite the lack of crowd during the COVID-19 pandemic season (2019/2020), which could have affected the home advantage effect (14), there was no noticeable difference in the home vs. away team formation dynamic in comparison with the other seasons. It is worth noting that the two Italian teams examined in Figure 8 both won the Scudetto when using the same formation all season: Juve (3-5-2), record points and wins in 2013/2014 and AC Milan (4-5-1) in 2021/2022.

Effect of opponent's team formation

There were clear differences in some proportions of team formations used when playing against 3- vs. 4-defender formations (Figure 9), suggesting a kind of mirroring effect. In fact, and consistently across all leagues, coaches tended to use more 3-defender formations when playing against 3-defender formations, and more 4-5-1 (and to a lesser extent 4-2-2) formations when playing against 4-defender formations. This may be explained by different reasons, but playing with 3 defenders against a 3-defenders formation may help to block the oppo-

nent's full-backs (5 vs. 5 in the width of the pitch), not to face inferiority in the midfield. It is also likely easier to press when the number of players is even (man to man).

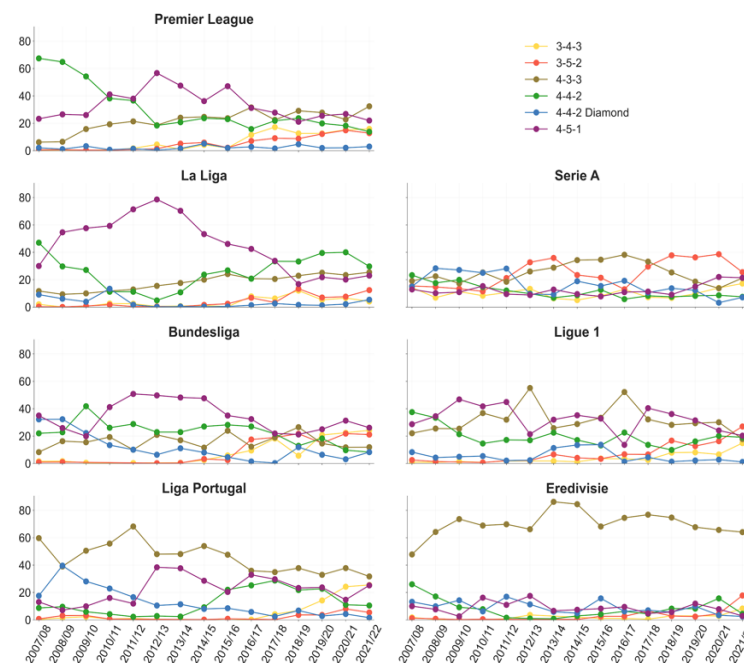


Fig. 6. Proportion of team formations for the seven leagues individually from 2007/08 to the 2021/22 seasons.

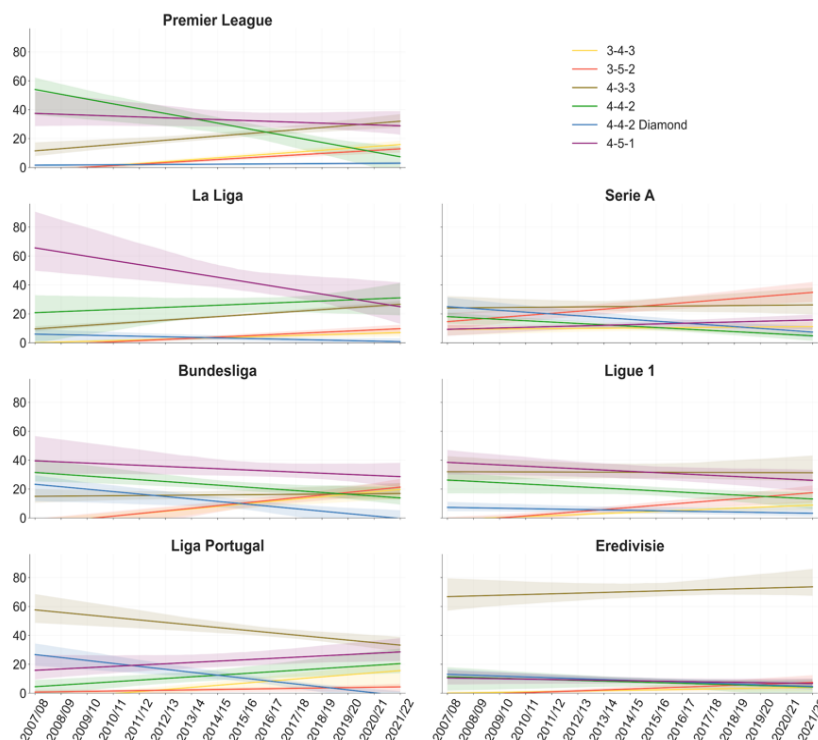


Fig. 7. Linear trends for the proportion of team formations for the seven leagues individually from 2007/08 to the 2021/22 seasons.

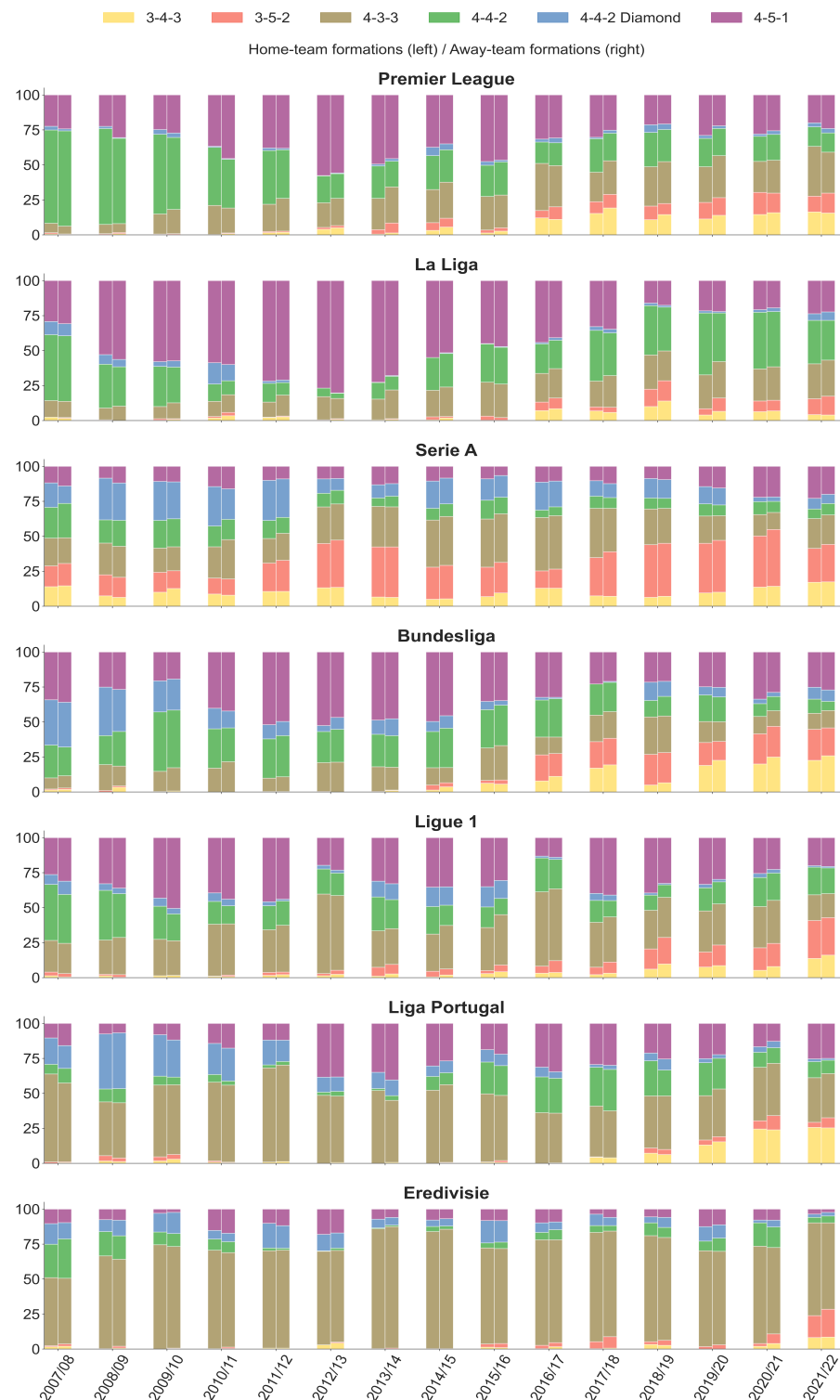


Fig. 8. Proportion of team formations used in each of the seven leagues as a function of match location from 2007/08 to the 2021/22 season (left: home, right: away matches).

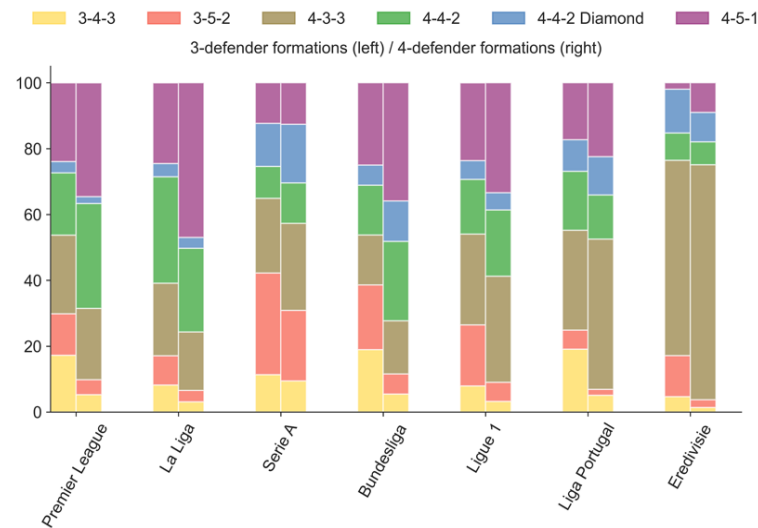


Fig. 9. Proportion of team formations used when playing against 3- and 4-defender formations in the seven leagues across the entire period analysed.

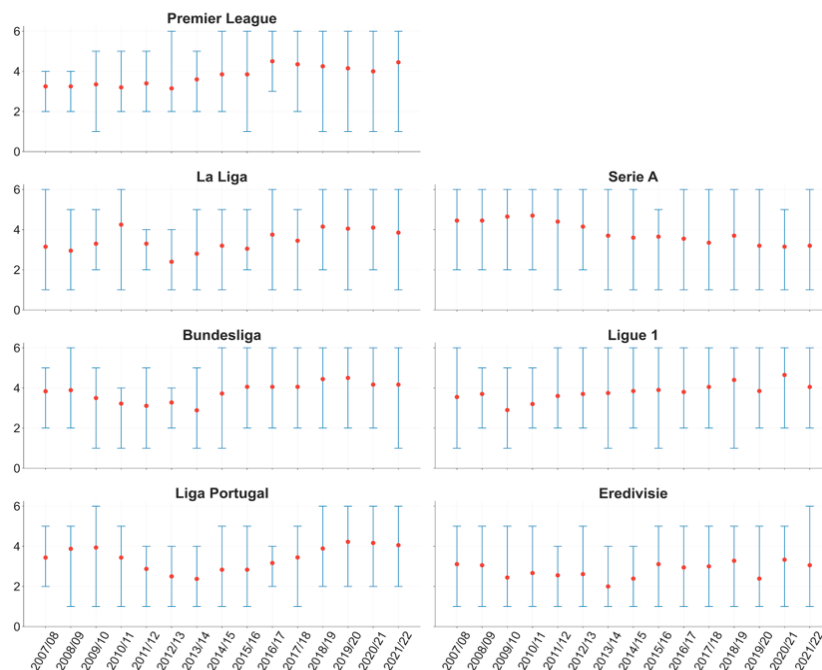


Fig. 10. Number of different team formations used by individual teams in the seven leagues from 2007/08 to the 2021/22 season.

Formation changes within teams at the league level

Beyond the first examination of the above-mentioned overall trends, we also looked at within-team variations in team formations. As shown in Figure 10, most teams were reported to use between 3 and 4.5 different formations on average within a single season (range: 1-6 in all leagues). While there was an increasing trend for greater variability in team formations in the EPL over time (consistently <4 before 2013/2014, and then consistently >4 , Figure 1), the number of team formations used in the Serie A tended to decrease across the years

(>4 up to 2012/2013 and no more than 3 since 2019/2020). Interestingly, however, teams still tended to predominantly use one system over the other (Figure 11). The percentage of use of the preferred system ranged from 50% to 90%, depending on the teams, the league and the years. In the Eredivisie, for example, the main team formation was used 80% to 90% of the time, suggesting that changes may only be very transient and targeted at some specific matches/contexts. In the French Ligue 1 in contrast, the preferred team formation was used only 50% to 60% of the time, reflecting a greater variability and flexibility in the tactical strategies of coaches, who

may change regularly their formations based on the weekly opposition and context (i.e., match location).

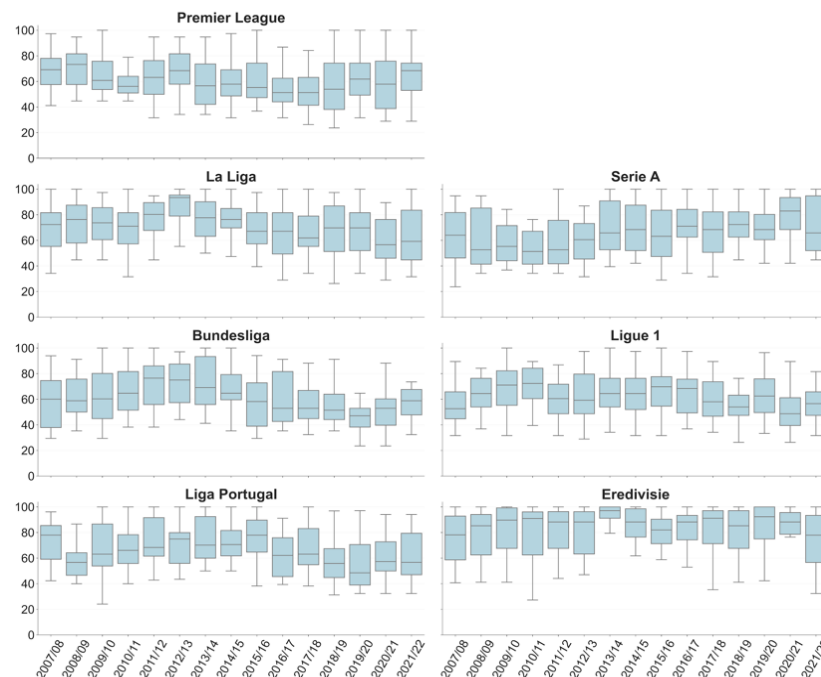


Fig. 11. Proportion of use of the more frequent team formation for each individual team in the seven leagues from 2007/08 to the 2021/22 season.

Formation changes within a selection of top European teams

Finally, the specific case studies of top European teams (Figures 12 and 13) showed how versatile and variable the use of those systems is at the club level - whether it's looking within a season, across a manager cycle, and throughout the entire period analysed. Following transient changes in team formation to support specific tactics when playing a given opponent or when playing home vs. away (Figure 8), the greater driver for changes in team formation is obviously managers' preferences and style of play. Almost all examples in our case studies (Figures 12 and 13, and Tables 2 and 3) support this manager's influence.

For example in Man City (Figure 12, Table 2), Roberto Mancini played mainly in 4-5-1 (2010/2011 to 2012/2013) after a transition year when he joined mid-season in 2009/2010 (Mark Hughes had been using the 4-3-3 from 2008/09 to the first part of 2009/2010). When Manuel Pellegrini started to coach in 2013/2014, he used the 4-4-2 more often; a formation that was almost completely abandoned in favour of the 4-3-3 until now when Pep Guardiola joined in 2016/2017. Interestingly however, despite the 4-3-3 being the dominant formation across the season, Man City still played sometimes with 3-defender formations against top teams like Chelsea, Liverpool, and Arsenal.

At Manchester United (Figure 13, Table 3), the dynamic reflected Sir Ferguson's preferences for the 4-4-2 (with parallel use of the 4-5-1, from 10% to 40%, depending on the year) until 2011/2012, before he changed to favour the 4-5-1 (80% at home) during his last season (2012/2013). It is interesting to note during the 2007/2008 and 2008/2009 that United were reported to prefer 4-5-1 and 4-4-2 which may not fully reflect the flexibility of the players within the squad at the

time. Particularly with forwards Cristiano Ronaldo, who was transitioning from a winger to central forward, Wayne Rooney and Carlos Tevez often creating a forward line of three and interchanging positions during defensive and offensive phases of play.

The 4-5-1 was also used in similar proportions by David Moyes during the consecutive, post-Ferguson season (2013/2014). Interestingly, however, when Louis Van Gaal joined in 2014/2015, he used four (home) to five (away) different team formations at almost similar proportions; whether it was more a willingness to adapt to each opponent, versus a recurring process to find the best playing options in a club that was in a difficult situation at the start of the season (i.e., after 10 league matches, United were in ninth place with 13 points and three victories, their worst start to the season since 1986–87, before finally finishing fourth) remains difficult to answer with the data available.

As per the experience of one of the co-authors, the fact that PSG used heavily the 4-3-3 between 2013/2014 and 2017/2018 (Figure 13, precisely initiated under Laurent Blanc and continued by Unai Emery, since Carlo Ancelotti wasn't using this system the years before) was related to a mix between coaches' preferences and more importantly, players' characteristics (Matuidi/Motta/Verratti being considered at that time one of the best midfield in Europe). A 4-5-1 was however sometimes used against specific teams, such as Olympique Lyonnais, against whom they always had highly contested matches during the past 10 years. The arrival of Thomas Tuchel in 2018/2019, known for his orientation toward flexible systems of play is well supported by the use of a variety of systems (all used in an almost similar manner except the 4-4-2 Diamond), especially during his first season. As per the example of Louis Van Gaal at Manchester United in 2014/2015, it is still under debate whether this was more a process to find the best play-

ing options with such a large squad of talent or a clear strategy to adapt to each opponent.

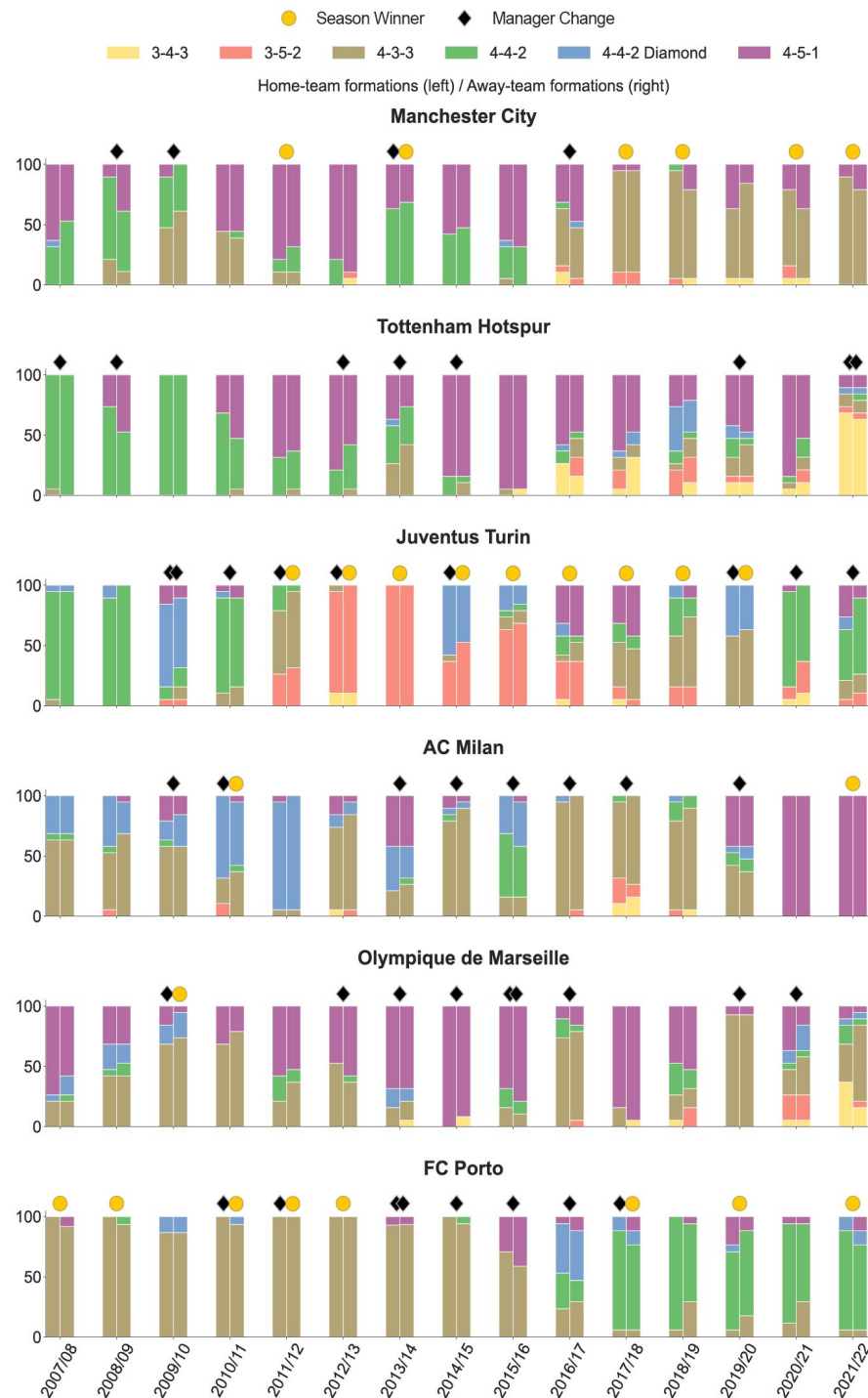


Fig. 12. Proportion of team formations used by selected top teams as a function of match location from 2007/08 to the 2021/22 season (left: home, right: away matches). Black diamonds indicate a change in manager during the given season (more details provided in Table 2); yellow circles indicate winning the league that same season.

Club cultures and team formation stability

Within the selected teams, Porto FC and FC Barcelona stand out for the fact that during long periods of time, the same (and almost unique) system was used over years. At FC Porto, the 4-3-3 was used almost exclusively up to 2015/2016 by six consecutive coaches (Jesualdo Ferreira, Andre Villas-Boas, Vitor Pereira, Paulo Fonseca, Luis Castro and Julen Lopetegui), before being abandoned in favour of the 4-5-1 in 2015/2016 with the arrival of Jose Peseiro, and then the 4-4-2 and 4-4-2 Diamond under Nuno Espirito Santo and Sergio Conceicao (Figure 12). This team formation stability was associated with one of the most successful periods of the club (winning five titles out of six seasons from 2007/08 to 2012/13). The drop in results between 2013/14 to 2015/16 may have therefore called for a change (in staff at least, and likely more changes also at the club and player level, which can't be commented on in the present manuscript), and was then followed by another successful period using the 4-4-2 until now (i.e., they won three titles and finished second three times from 2016/17 to 2021/22). At FC Barcelona, there was also a very important culture of the 4-3-3 (under six different managers, including Frank Rijkaard, Pep Guardiola, Tito Vilanova, Gerardo Martino and Luis Enrique), until 2017/2018 when Ernesto Valverde introduced the 4-4-2 for the first time since (at least) 2007/08. Interestingly, Pep Guardiola and Luis Enrique sometimes also used the 3-4-3 (25% of matches) but exclusively during the 2011/2012 and 2016/2017 seasons, respectively. This likely reflects specific approaches of a few particular matches, for which the 3-4-3 formation was thought to be more effective than the 4-3-3, with no clear relationship with the opponent team status, formation and outcome (e.g., during the 2011/2012 season, FCB lost at home vs. Real Madrid, 4-5-1, 1st, drew away vs. Valencia, 4-5-1, 3rd, and won both at home and away vs. Levante, 4-3-3 6th; Osasuna, 4-3-3; 7th, Mallorca 4-4-2, 8th; Betis Sevilla 4-3-3, 13th; and Villareal, 4-4-2, 18th). Similar to the above, Liverpool swapped its 4-3-3 for a 4-5-1 during the 2019/2020 EPL season-winning for four matches. All those matches occurred in December - a highly congested month - against teams supposedly weaker, with the aim of resting at least one player of their famous Front Three (Roberto Firmino,

Sadio Mané and Mohamed Salah): Liverpool won all matches vs. FC Everton, 3-4-3, 12th; Wolverhampton, 3-4-3 7th; and AFC Bournemouth, 4-4-2, 18th; Watford, 4-5-1, 19th.

Interestingly, some successive managers within a single season may also still use the same formation; whether it is because they shared the same preference, or didn't want to disturb too many players' habits is difficult to answer. At Juventus, Antonio Conte started using 4-3-3 in 2011/12 before switching to 3-5-2 due to not being satisfied enough with the left-back position (8) and having three elite centre-backs (Andrea Barzagli, Leonardo Bonucci and Giorgio Chiellini). He then fully adopted this formation during the next two successful seasons. This was the start of Conte's interest in 3-defender formations, especially thanks to their wing-backs contribution, as he has been mainly using 3-4-3 at Tottenham since 2021/22.

Finally, there does not seem to be an association between formation stability at season level, i.e., when using one formation almost 100% of the time during a given season, and overall season performance, i.e., League title (Figures 12 and 13). For instance, while Juventus won the 2012/13 and 2013/14 Serie A titles using almost exclusively the 3-5-2, they won again the following six titles while varying substantially their systems within the same season (with for example all possible systems used in 2016/2017). This could have been explained by the fact that Juventus won those titles by a significant margin with their closest opponents, but this was not the case in 2016/17 (4 points from AS Roma), 2017/18 (4 points from Napoli) and 2019/20 (1 point from Inter Milan). This formation variability across successive successful seasons was also the case for Real Madrid's 2007/08, Manchester United's 2007/08 and 2010/11, Barcelona's 2017/18, Bayern Munich's 2014/15 and 2015/16, and Paris' 2018/19 and 2019/20 squads (Figure 13).

There also does not seem to be an association between formation stability at club level, i.e., the use of the same formation over time, and overall performance. For instance, FC Porto mainly used 4-3-3 during a highly successful cycle, i.e., five titles in six years between 2007/08 and 2012/13 (Figure 12) and went on to win again in 2017/18, 2019/20 and 2021/22 mainly using 4-4-2 instead of 4-3-3.

Table 2. List of managers in charge of the six teams selected in Figure 12, together with dates and duration of service. Services shorter than two months are not shown.

Team	Manager	Season	Start Date	End Date	Duration
Manchester City	sven-goran-eriksson	2007/08	2007-08-11	2008-05-11	274
	mark-hughes	2008/09 2009/10	2008-08-17	2009-12-19	489
	roberto-mancini	2009/10 2010/11 2011/12 2012/13	2009-12-26	2013-05-19	1240
	manuel-pellegrini	2013/14 2014/15 2015/16	2013-08-19	2016-05-15	1000
	pep-guardiola	2016/17 2017/18 2018/19 2019/20 2020/21 2021/22	2016-08-13	2022-05-22	2108

Tottenham Hotspur	martin-jol	2007/08	2007-08-11	2007-10-22	72
	juande-ramos	2007/08 2008/09	2007-10-28	2008-10-19	357
	harry-redknapp	2008/09 2009/10 2010/11 2011/12	2008-10-26	2012-05-13	1295
	andre-villas-boas	2012/13 2013/14	2012-08-18	2013-12-15	484
	tim-sherwood	2013/14	2013-12-22	2014-05-11	140
	mauricio-pochettino	2014/15 2015/16 2016/17 2017/18 2018/19 2019/20	2014-08-16	2019-11-09	1911
	jose-mourinho	2019/20 2020/21	2019-11-23	2021-04-16	510
	nuno-espírito-santo	2021/22	2021-08-15	2021-10-30	76
	antonio-conte	2021/22	2021-11-07	2022-05-22	196
Juventus Turin	claudio-ranieri	2007/08 2008/09	2007-08-25	2009-05-17	631
	ciro-ferrara	2008/09 2009/10	2009-05-24	2010-01-23	244
	alberto-zaccheroni	2009/10	2010-01-31	2010-05-15	104
	luigi-delneri	2010/11	2010-08-29	2011-05-22	266
	antonio-conte	2011/12	2011-09-11	2012-05-13	245
	antonio-conte	2012/13 2013/14	2012-12-09	2014-05-18	525
	massimiliano-allegri	2014/15 2015/16 2016/17 2017/18 2018/19	2014-08-30	2019-05-26	1730
	maurizio-sarri	2019/20	2019-08-24	2020-08-01	343
	andrea-pirlo	2020/21	2020-09-20	2021-05-23	245
AC Milan	carlo-ancelotti	2007/08 2008/09	2007-08-26	2009-05-31	644
	leonardo	2009/10	2009-08-22	2010-05-15	266
	massimiliano-allegri	2010/11 2011/12 2012/13 2013/14	2010-08-29	2014-01-12	1232
	clarence-seedorf	2013/14	2014-01-19	2014-05-18	119
	filippo-inzaghi	2014/15	2014-08-31	2015-05-30	272
	sini-scaron-a-mihajlovic	2015/16	2015-08-23	2016-04-09	230
	vincenzo-montella	2016/17 2017/18	2016-08-21	2017-11-26	462
	gennaro-gattuso	2017/18 2018/19	2017-12-03	2019-05-26	539
	stefano-pioli	2019/20 2020/21 2021/22	2019-10-20	2022-05-22	945

Olympic Marseille	eric-gerets	2007/08 2008/09	2007-10-06	2009-05-30	602
	didier-deschamps	2009/10 2010/11 2011/12	2009-08-08	2012-05-20	1016
	elie-baup	2012/13 2013/14	2012-08-12	2013-12-06	481
	jose-anigo	2013/14	2013-12-15	2014-05-17	153
	marcelo-bielsa	2014/15 2015/16	2014-08-09	2015-08-08	364
	michel	2015/16	2015-08-23	2016-04-17	238
	franck-passi	2015/16 2016/17	2016-04-24	2016-10-16	175
	rudi-garcia	2016/17 2017/18 2018/19	2016-10-23	2019-05-24	943
	andre-villas-boas	2019/20 2020/21	2019-08-10	2021-01-23	532
FC Porto	jorge-sampaoli	2020/21 2021/22	2021-03-10	2022-05-21	437
	jesualdo-ferreira	2007/08 2008/09 2009/10	2007-08-18	2010-05-08	994
	andre-villas-boas	2010/11	2010-08-14	2011-05-14	273
	vitior-pereira	2011/12 2012/13	2011-08-14	2013-05-19	644
	paulo-fonseca	2013/14	2013-08-18	2014-03-02	196
	luis-castro	2013/14	2014-03-09	2014-05-10	62
	julen-lopetegui	2014/15 2015/16	2014-08-15	2016-01-17	520
	jose-peseiro	2015/16	2016-01-24	2016-05-14	111
	nuno-espirito-santo	2016/17	2016-08-12	2017-05-21	282
	sergio-conceicao	2017/18 2018/19 2019/20 2020/21 2021/22	2017-08-09	2022-05-14	1739

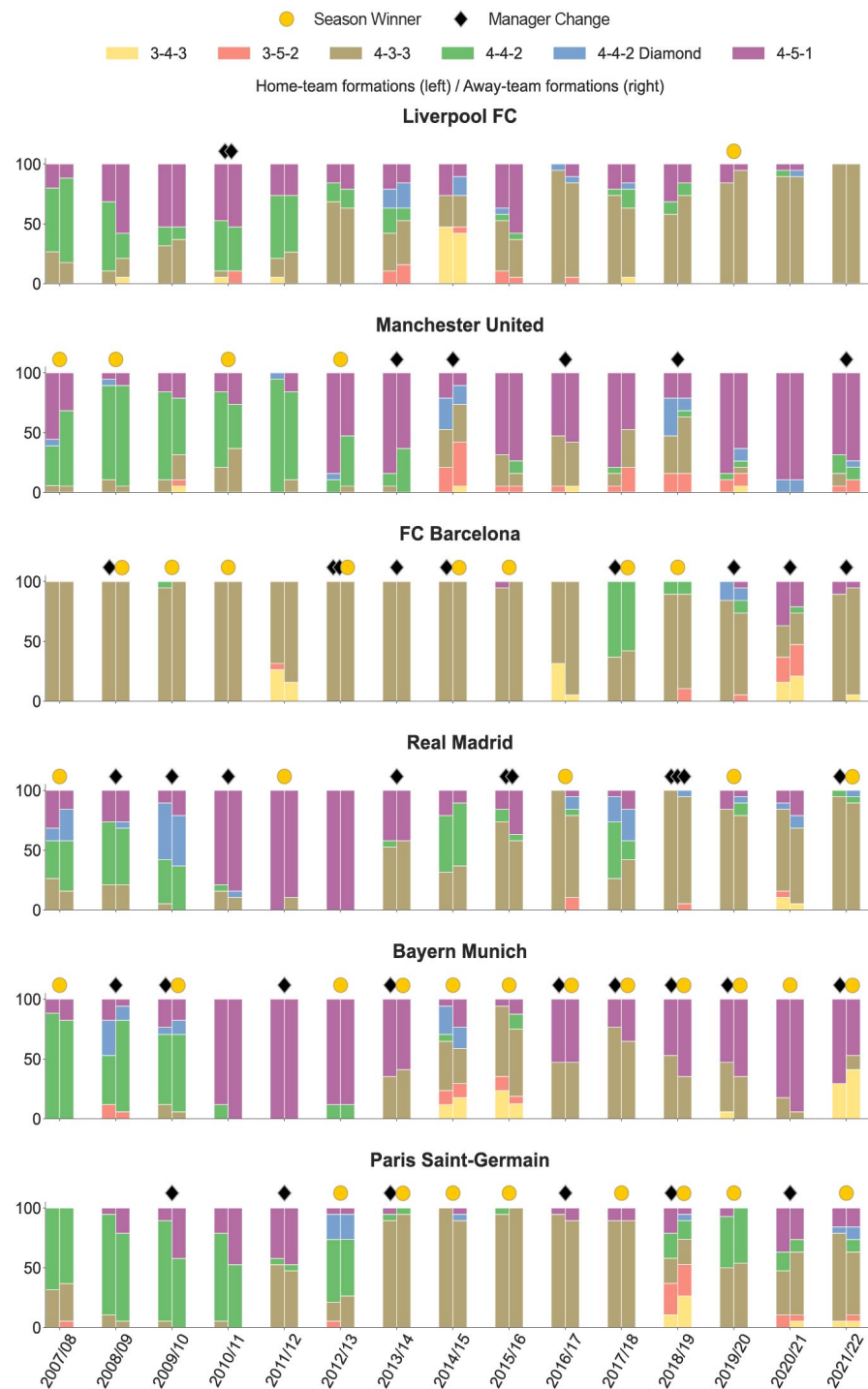


Fig. 13. Proportion of team formations used by selected top teams as a function of match location from 2007/08 to the 2021/22 season (left: home, right: away matches). Black diamonds indicate a change in manager during the given season (more details provided in Table 3); yellow circles indicate winning the league that same season.

Table 3. List of managers in charge of the six teams selected in Figure 13, together with dates and duration of service. Services shorter than two months are not shown.

Team	Manager	Season	Start Date	End Date	Duration
FC Liverpool	rafael-benitez	2007/08 2008/09 2009/10	2007-08-11	2010-05-09	1002
	roy-hodgson	2010/11	2010-08-15	2011-01-05	143
	sir-kenny-dalglish	2010/11 2011/12	2011-01-12	2012-05-13	487
	brendan-rodgers	2012/13 2013/14 2014/15 2015/16	2012-08-18	2015-10-04	1142
	jurgen-klopp	2015/16 2016/17 2017/18 2018/19 2019/20 2020/21 2021/22	2015-10-17	2022-05-22	2409
Manchester United	sir-alex-ferguson	2007/08 2008/09 2009/10 2010/11 2011/12 2012/13	2007-08-12	2013-05-19	2107
	david-moyes	2013/14	2013-08-17	2014-05-11	267
	louis-van-gaal	2014/15 2015/16	2014-08-16	2016-05-17	640
	jose-mourinho	2016/17 2017/18 2018/19	2016-08-14	2018-12-16	854
	ole-gunnar-solskjaer	2018/19 2019/20 2020/21 2021/22	2018-12-22	2021-12-02	1076
	ralf-rangnick	2021/22	2021-12-05	2022-05-22	168
FC Barcelona	frank-rijkaard	2007/08	2007-08-26	2008-05-17	265
	pep-guardiola	2008/09 2009/10 2010/11 2011/12	2008-08-31	2012-05-12	1350
	tito-vilanova	2012/13	2012-08-19	2012-12-16	119
	jordi-roura	2012/13	2013-01-27	2013-03-30	62
	gerardo-martino	2013/14	2013-08-18	2014-05-17	272
	luis-enrique	2014/15 2015/16 2016/17	2014-08-24	2017-05-21	1001
	ernesto-valverde	2017/18 2018/19 2019/20	2017-08-20	2020-01-04	867
	quique-setien	2019/20	2020-01-19	2020-07-19	182
	ronald-koeman	2020/21 2021/22	2020-09-27	2021-11-06	405
	xavi	2021/22	2021-11-20	2022-05-22	183

Real Madrid	bernd-schuster	2007/08 2008/09	2007-08-25	2008-12-07	470
	juande-ramos	2008/09	2008-12-13	2009-05-31	169
	manuel-pellegrini	2009/10	2009-08-29	2010-05-16	260
	jose-mourinho	2010/11 2011/12 2012/13	2010-08-29	2013-06-01	1007
	carlo-ancelotti	2013/14 2014/15	2013-08-18	2015-05-23	643
	rafael-benitez	2015/16	2015-08-23	2016-01-03	133
	zinedine-zidane	2015/16 2016/17 2017/18	2016-01-09	2018-05-19	861
	julen-lopetegui	2018/19	2018-08-19	2018-10-28	70
	santiago-solari	2018/19	2018-11-03	2019-03-10	127
	zinedine-zidane	2018/19 2019/20 2020/21	2019-03-16	2021-05-22	798
FC Bayern Munchen	carlo-ancelotti	2021/22	2021-08-14	2022-05-20	279
	ottmar-hitzfeld	2007/08	2007-08-11	2008-05-17	280
	jurgen-klinsmann	2008/09	2008-08-15	2009-04-25	253
	louis-van-gaal	2009/10 2010/11	2009-08-08	2011-05-14	644
	jupp-heyckes	2011/12 2012/13	2011-08-07	2013-05-18	650
	pep-guardiola	2013/14 2014/15 2015/16	2013-08-09	2016-05-14	1009
	carlo-ancelotti	2016/17 2017/18	2016-08-26	2017-10-01	401
	jupp-heyckes	2017/18	2017-10-14	2018-05-12	210
	niko-kovac	2018/19 2019/20	2018-08-24	2019-11-02	435
	hansi-flick	2019/20 2020/21	2019-11-09	2021-05-22	560
Paris-saint-germain	julian-nagelsmann	2021/22	2021-08-13	2022-05-14	274
	paul-le-guen	2007/08 2008/09 2020/21	2007-08-04	2009-05-30	665
	antoine-kombouare	2009/10 2010/11 2011/12	2009-08-08	2011-12-21	865
	carlo-ancelotti	2011/12 2012/13	2012-01-14	2013-05-26	498
	laurent-blanc	2013/14 2014/15 2015/16	2013-08-09	2016-05-14	1009
	unai-emery	2016/17 2017/18	2016-08-12	2018-05-19	645
	thomas-tuchel	2018/19 2019/20 2020/21	2018-08-12	2020-12-23	864
	mauricio-pochettino	2020/21 2021/22	2021-01-06	2022-05-21	500

Overall, those case studies show clearly that except in some rare cases (e.g., FC Porto and FC Barcelona during some years), there haven't been club-specific styles of play - most of the time, the main preferred formations rotated by cycles, often with the arrival of a new manager (and in parallel, squad rotation). The main challenge for managers remains which team formation to choose in a given context, balancing their own preferences and biases, club context and history, players' profiles, strengths and availability, and of course, the opponent and the location of a given fixture.

Key findings

- The 4-3-3 (31%), 4-5-1 (28%) and 4-4-2 (19%) were the most used systems over the 15-year period.
- There was a clear trend for a progressive decrease in the use of the 4-4-2 (including 4-4-2 Diamond shape) and 4-5-1 (including 4-2-3-1) over time, with teams shifting progressively to using more 3-defenders systems (i.e., 3-4-3, 3-5-2).
- Except in the Eredivisie, where there is a very strong 4-3-3 culture, there was no clear trend for a given system to really stand out in comparison to others when looking at all leagues together.
- There have been many cycles over the years in almost every league for a given formation to dominate over seven to nine consecutive years before coaches started to favour another (trial/error, trends, etc.).
- Over the past five to six seasons, there has been a tendency for a greater uniformisation of the tactical approach to matches across leagues, with a less and less marked preference for a formation vs. another.
- Except for a few exceptions, there doesn't seem to exist tactical club cultures either, with preferred formations often changing as new coaches and managers come in.
- The decision to use one formation over another may result from the combination of several factors including league and club culture (likely moderate and limited influence, respectively), coaches' experiences both as ex-players and managers, overall tactical preferences and philosophy, and fixture context (opponent, players' availability, timing and location).
- Overall, formations should be adjusted to the (in)abilities of players rather than the players adapting to a formation.
- Following the above, some coaches use up to six formations within the same season.
- There doesn't seem to be any association between team formations (and especially the stability of a formation) and success across all leagues and clubs examined (at least data limited to the present case studies).
- Overall, this suggests that players need to develop their ability to adapt and compete in almost all possible formations, which may change from a club to another, and/or with a manager or another (and many changes can happen within a single season).
- The most tactically adaptable players may target/move to any league, while the less adaptable ones may target the leagues with low changes (i.e., Eredivisie playing mostly in 4-3-3).

Acknowledgements

The authors thank Willian Mendes for his work on the figures.
Twitter: Martin Buchheit (@mart1buch), Adrien Tarascon (@TarasconAdrien), Karim Hader (@Karad70), Derek McHugh (@DerekMcHugh3), Raymond Verheijen (@raymondverheije)

References

1. Alberti, G., Iaia, F. M., Arcelli, E., Cavaggioni, L., and Rampinini, E. (2013). Goal scoring patterns in major European soccer leagues. *Sport Sci. Health* 9, 151–153. doi: 10.1007/s11332-013-0154-9
2. Buchheit M, Settembre M, Hader K, Tarascon A, McHugh D & Verheijen R. Do mid-week European matches influence European teams' performance in their domestic league? A 20-year study. *Sport Perf & Science Reports*, Oct 22, 175, v1
3. Crolley, L., Hand, D., and Jeutter, R. (2000). Playing the identity card: stereotypes in European football. *J. Soccer Soc.* 1, 107–128. doi: 10.1080/14660970008721267
4. Defrance, J. *Sociologie du Sport*, 2009. Repères, La découverte.
5. Dellal, A., Chamari, K., Wong, D. P., Ahmaidi, S., Keller, D., Barros, R., et al. (2011). Comparison of physical and technical performance in European soccer match-play: FA premier league and La Liga. *Eur. J. Sport Sci.* 11, 51–59. doi: 10.1080/17461391.2010.481334
6. Lago-Peñas, C., Gómez-Ruano, M., Megías-Navarro, D., and Pollard, R. (2016). Home advantage in football: examining the effect of scoring first on match outcome in the five major European leagues. *Int. J. Perform. Anal. Sport* 16, 411–421. doi: 10.1080/24748668.2016.11868897
7. Liu, H., Yi, Q., Giménez, J.-V., Gómez, M.-A., and Lago-Peñas, C. (2015). Performance profiles of football teams in the UEFA champions league considering situational efficiency. *Int. J. Perform. Anal. Sport* 15, 371–390. doi: 10.1080/24748668.2015.11868799
8. Lopresti S. Juventus: Evaluating the Era of Antonio Conte's Bianconeri. *Bleacherreport blog*, July 11, 214.
9. Pollard R. Home advantage in soccer: a retrospective analysis. *J Sports Sci* 1986 Winter;4(3):237-48. doi: 10.1080/02640418608732122.
10. Poli R, Ravenel L and Besson R. Player turnover strategies in the five major European leagues. *CIES Football Observatory Monthly Report n°38 - October 2018*
11. Poli R, Ravenel L and Besson R. Performance and playing styles in 35 European football leagues. *CIES Football Observatory Monthly Report n°28 - October 2017*
12. Pundit Arena Team. Is The Traditional 4-4-2 Formation Outdated In Modern Football? Sep 12, 2018
13. Sarmiento, H., Pereira, A., Matos, N., Campaniço, J., Anguera, T. M., and Leitão, J. (2013). English Premier League, Spain's La Liga and Italy's Serie A—What's different? *Int. J. Perform. Anal. Sport* 13, 773–789.
14. Sors F, Grassi M, Agostini T, Murgia M. A complete season with attendance restrictions confirms the relevant contribution of spectators to home advantage and referee bias in association football. *PeerJ.* 2022 Jul 4;10:e13681. doi: 10.7717/peerj.13681. eCollection 2022.

15. Verheijen R. Tactical principles. Methodological steps within tactical principles during team building process. Football Coach Evolution. June 2022.
16. Wilson J. The Question: What have been the tactical lessons of World Cup 2010? The Guardian, July 2010
17. Yi Q, Groom R, Dai C, Liu H and Gómez Ruano MÁ (2019) Differences in Technical Performance of Players From ‘The Big Five’ European Football Leagues in the UEFA Champions League. Front. Psychol. 10:2738. doi: 10.3389/fpsyg.2019.02738

syg.2019.02738

Copyright: The article published on Science Performance and Science Reports are distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution and reproduction in any medium, provided you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made. The Creative Commons Public Domain Dedication waiver (<http://creativecommons.org/publicdomain/zero/1.0/>) applies to the data made available in this article, unless otherwise stated.

